DSNet: Dynamic Stitchable Neural Network for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image classification (HSIC) aims to identify land-cover categories by leveraging the spectral and spatial information contained in hyperspectral images (HSIs). Currently, many deep learning approaches utilize dual-branch networks to process spectral and spatial data separately, followed by the application of specialized modules to facilitate feature interaction or fusion. However, the design of these modules demands considerable time and effort from researchers and may not adequately capture the inherent relationships between independent spatial and spectral features dynamically. To address these issues, we propose the dynamic stitchable neural network (DSNet) for HSIC. While the DSNet maintains a dual-branch structure, it operates without traditional feature fusion or interaction. Instead, it employs a stitching network approach to integrate the two branches. Specifically, a spatial–spectral stitching module is presented to incorporate multiple stitching layers at various positions between the two network branches, creating new stitched networks that retain the strengths of both original networks. Additionally, a reinforcement learning-based strategy (RLBS) is designed for dynamically selecting stitching positions tailored to specific datasets, enabling the model to adaptively optimize the integration of spatial and spectral features. Recognizing the effectiveness of the Vision Transformer (ViT) in learning spatial information and the capability of a 1-D convolutional neural network (1DCNN) in capturing spectral details, the DSNet directly stitches these two networks together. This fusion maximizes the utilization of both foundational networks, yielding a new hybrid network that delivers exceptional performance while also alleviating the burden on researchers to develop new architectures from scratch. Extensive experiments and analyses conducted on three public HSI datasets demonstrate the superiority of the proposed method, validating the effectiveness of our innovative modules. The codes of this work will be available at https://github.com/ZZC/IEEE-TGRS-DSNet**

***Index Terms*— Dynamic neural network, hyperspectral image classification (HSIC), reinforcement learning, stitching network.**

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I. Introduction

HYPERSPECTRAL images (HSIs) contain rich spatial and spectral information, which are valuable for land-cover identification [1]. Therefore, HSIs are widely applied in various fields such as agriculture, urban planning, and environmental monitoring [2]. The goal of HSI classification (HSIC) is to assign different semantic labels to each pixel in the HSI [3], which is a fundamental, yet challenging task in this field [4]. In recent years, the development of deep learning technology has led to higher accuracy in HSIC [5] due to its ability to provide end-to-end feature extractors and classifiers with global optimization. Therefore, most advanced methods in this field are currently based on deep learning methods.

In the earlier stages, HSIC models based on deep learning usually directly leveraged foundational network architectures such as the 1-D convolutional neural network (1DCNN) [6], 2-D convolutional neural network (2DCNN) [7], 3-D convolutional neural network (3DCNN) [8], and Vision Transformer (ViT) [9]. Due to their relatively simple structure and data-processing methods, these models cannot effectively learn and balance spatial and spectral knowledge simultaneously, resulting in limited performance improvement.

As the field continues to evolve, a growing body of advanced methodologies has emerged, bringing increased diversity to both neural network architectures and data-processing paradigms. For instance, Yao et al. [10] introduced a pseudolabel-based unreliable sample learning (PUSL) framework to exploit low-confidence unlabeled data for semi-supervised HSIC. Xu et al. [11] proposed an orientational clustering learning (OCL) strategy leveraging class anchors

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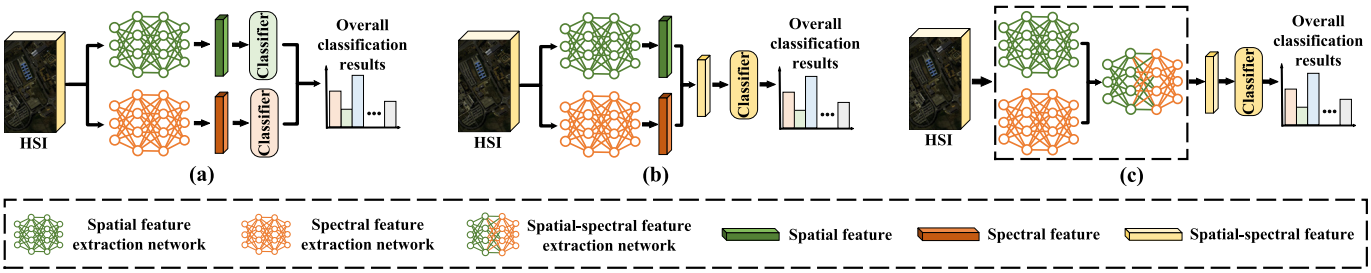


Fig. 1. Workflow of integrating spatial knowledge and spectral knowledge from different levels. (a) Integrating spatial information and spectral information at the level of classification results. (b) Fusing spatial information and spectral information from the perspective of intermediate network features. (c) Directly combining two network branches to integrate spatial knowledge and spectral knowledge at the network level.

to enhance interclass separability and reduce intraclass feature distance. Zhang et al. [12] designed an innovative non-Euclidean spectral–spatial feature mining network combining the gated graph convolutional network (GCN) and CNN (NESSGGCN) to jointly extract features from the non-Euclidean domain via weighted fusion.

Among these advanced methodologies for HSIC based on deep learning, a substantial portion of them aimed to capture both spatial and spectral information to learn more comprehensive knowledge from HSIs. While some approaches, such as the 3DCNN, attempted to jointly extract spectral–spatial features by using a single network branch, they often struggled to balance the contributions from each modality [13]. To address this, many subsequent studies adopted dual-branch architectures [14], which independently learn spatial and spectral features and fuse them via concatenation [15], cross-branch interactions [16], or postbranch modules [17]. Such frameworks provide enhanced flexibility in each network branch design, enabling richer feature extraction and improved modality balance [18]. However, they often fall short in modeling nonlinear relationships between modalities [19], require substantial manual design of fusion mechanisms [20], and may inadequately capture intrinsic correlations [21]. Moreover, most methods rely on fixed architectures with full-parameter training, which limits generalization and causes performance inconsistency across datasets [22].

From the above analysis, it can be seen that most existing HSIC methods integrate spatial and spectral features either at the classification level [23] [Fig. 1(a)] or via intermediate network features [24] [Fig. 1(b)], while largely neglecting direct fusion at the network level [Fig. 1(c)]. Obviously, network-level integration enables more flexible design of each network branch to balance the contribution between spatial and spectral knowledge and facilitates end-to-end learning of enriched spectral–spatial representations [25], which offers a promising solution to address existing limitations. It is noteworthy that many stitching network methods have recently emerged as an effective paradigm in the field of computer vision, enabling the fusion of heterogeneous pretrained models by inserting lightweight stitching layers [26]. When applied to HSIC, this approach allows spatial and spectral branches to be directly combined at the network level, leveraging the strengths of existing well-designed foundational models and integrating the best parts of them to form a new one for improved performance. Such stitching frameworks can not only offer improved performance but also significantly save time and effort in developing specific improvement strategies [27].

However, as stitching positions between two network branches can be arbitrarily defined, determining optimal positions for specific HSI inputs is nontrivial. Different stitched locations lead to distinct network behaviors, and optimal positions may vary across datasets. To address this issue, dynamic neural network technology can be used to dynamically select networks generated by different stitch positions [22]. This technology has now developed very maturely, among which the most representative ones include early reinforcement learning methods and currently widely used neural architecture search (NAS) methods [28]. While NAS methods have been widely adopted for architecture selection, they are often tailored for complex search spaces and Pareto-efficient tradeoffs [29], making them suboptimal for stitching-based super-networks, which are typically more constrained and structurally simple [30]. However, from the research [31], [32], it can be seen that some early reinforcement learning-based studies are relatively more suitable for the dynamic selection of stitch positions in this stitch architecture.

To address all these issues, this article proposes a novel dynamic stitchable neural network (DSNet) for HSIC. To directly integrate spatial and spectral pipelines at the network level, we employ the stitching method in the Snnet [30] to stitch two networks containing spatial and spectral knowledge, forming a spatial–spectral stitched super-network. Moreover, to select the appropriate stitched positions for different input HSIs, a reinforcement learning-based strategy (RLBS) is designed, and a stitching position selection network is introduced to assist in the selection of stitching positions (subnetworks) during the training process. Through the training strategy of reinforcement learning, both the position selection network and the stitching super-network can be optimized. Finally, two foundational networks are chosen to learn spatial and spectral knowledge, respectively, and are then stitched together to form several new networks. This can make better use of the foundational networks to create new networks with superior performance, while simultaneously saving the time and effort of the researchers in developing new networks.

In summary, the main contributions of this work can be outlined as follows.

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1) We introduce the concept of stitched networks into HSIC models and propose a novel DSNet. Because this method can directly stitch heterogeneous architecture networks together, the advantageous parts of the two foundational networks can be directly stitched together, which are used for extracting spatial and spectral information. To the best of our knowledge, this is the first attempt to apply the network stitching methods to the HSIC field.
2) We design a reinforcement learning-based training strategy, utilizing a stitching position selection network to dynamically select the appropriate stitched positions for the input data. This approach allows the network structure (i.e., stitching positions) to be flexibly adjusted based on the input data, significantly enhancing overall adaptability and generalization capabilities of the whole network architecture in this method.
3) Recognizing that the shallow layers of the ViT network efficiently encapsulate global spatial knowledge, while the deeper layers of the 1DCNN network acquire more intricate spectral information, we stitch the ViT and 1DCNN networks together to form a stitched super-network. Each newly generated stitched network not only incorporates the advantageous structures and parameters from both original networks but also directly learns comprehensive spatial–spectral features.

The remainder of this article is organized as follows. Section II provides a brief overview of network stitching. Section III offers a detailed description of the proposed DSNet. Section IV presents and analyzes the experimental results. Finally, Section V concludes this work.

II. Related Work

This section will provide a brief introduction to the network stitching technology. The concept of network stitching was initially proposed by the study in [33], where they employed a 1×1 convolutional stitching layer to connect the early layers of one pretrained network with the final layers of another pretrained network, thereby creating a new stitched network [34].

Currently, there are primarily two research pathways in the study of stitching neural networks. Some network stitching methods attempts to stitch networks from the perspective of network similarity representation, utilizing methods such as linear regression [35], canonical correlation analysis (CCA) [36], centered kernel alignment (CKA) [37], and generalized shape metrics (GSM) [38]. These studies have provided practical tools for the stitching and reconfiguration of models. For instance, Yang et al. [39] recently introduced the DeRy, which leverages the similarity between neural representations, and this approach allows for the reassembly of arbitrary pretrained models into a new network. However, the study in [26] indicated that the similarity index is not necessarily correlated with the performance of the stitched model.

Other network stitching methods directly attempt to use networks of different structures to stitch and explore the performance of newly generated networks at different positions from the perspective of network structure. For example, Bansal et al. [27] demonstrated that networks with varying structures or training strategies can be stitched with negligible accuracy loss. Steiner et al. [40] achieved competitive performance by grafting a ViT onto a ResNet. Building upon this concept, the stitchable neural network (Snnet) framework [30] introduces multiple learnable stitching layers between heterogeneous backbones, enabling the reuse of existing networks to dynamically generate multiple hybrid architectures. Given its ability to flexibly combine spatial and spectral branches and to automatically select multiple stitching positions simultaneously, the Snnet is well-suited for HSI modeling. Accordingly, this study adopts the Snnet paradigm to stitch the spatial and spectral branches of the network together for more effective HSIC.

III. Proposed Method

This section provides details of the DSNet. The overall training and test process of the proposed method is shown in Figs. 2 and 3. Its establishment process can be categorized into the construction of the following three components: the spatial–spectral network branch stitching module, the reinforcement learning strategy for selecting stitching positions, and the selection of spatial and spectral network structures.

For the convenience of subsequent theoretical exposition, the following notations are given. An HSI can be denoted as $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the height, width, and the number of bands of the HSI, respectively. Let the labeled samples in $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ be denoted as $\mathbf{X}^{\text{spec}} = \{\mathbf{x}_i^{\text{spec}}\}_i^{n_s} \in \mathbb{R}^{1\times 1\times B}$, where $\mathbf{x}_i^{\text{spec}}$ represents the ith pixel (spectral vector) in $\mathbf{X}^{\text{spec}}$. The corresponding class labels of $\mathbf{X}^{\text{spec}} = \{\mathbf{x}_i^{\text{spec}}\}_i^{n_s} \in \mathbb{R}^{1\times 1\times B}$ can be denoted as $\mathbf{Y}^s = \{\mathbf{y}_i^s\}_i^{n_s} \in \mathbb{R}^{1\times C}$, where n_s and C represent the size of the training sample and the number of classes in HSIs, respectively. If $\mathbf{x}_i^{\text{spec}} \in \mathbb{R}^{1\times 1\times B}$ is selected as the center pixel of a patch, $\mathbf{x}_i^{\text{spa}} \in \mathbb{R}^{p\times p\times B}$ can be defined as the hyperspectral patch around the center pixel $\mathbf{x}_i^{\text{spec}}$, where p represents the size of the spatial patch. The set formed by $\mathbf{x}_i^{\text{spa}}$ can be represented as $\mathbf{X}^{\text{spa}} = \{\mathbf{x}_i^{\text{spa}}\}_i^{n_s} \in \mathbb{R}^{p\times p\times B}$.

A. Spatial–Spectral Network Branch Stitching Module

This section will elaborate on the stitching process for spatial and spectral network branches. Assuming that there are two networks T_ϕ and H_θ, they are used to learn spatial information and spectral information knowledge from an HSI, respectively. Each network branch can be represented mathematically by manually dividing it into L layers. The function of the ith layer can be expressed as $f_{t,\phi,i}$ and $f_{h,\theta,i}$, and the entire network can be represented as

$$\begin{aligned} T_{\phi,1|L}\left(\mathbf{X}^{\text{spa}}\right) &= f_{t,\phi,L} \circ \cdots \circ f_{t,\phi,1}\left(\mathbf{X}^{\text{spa}}\right) = \mathbf{Y}^s \\ H_{\theta,1|L}\left(\mathbf{X}^{\text{spec}}\right) &= f_{h,\theta,L} \circ \cdots \circ f_{h,\theta,1}\left(\mathbf{X}^{\text{spec}}\right) = \mathbf{Y}^s \end{aligned} \tag{1}$$

where ϕ and θ represent the network parameters, $\circ$ denotes the composition, and L represents the network depth. $T_{\phi,1|L}: \mathbf{X}^{\text{spa}} \to \mathbf{Y}^s$ represents the network T_ϕ mapping the input $\mathbf{x}^{\text{spa}}$ from the input space $\mathbf{X}^{\text{spa}}$ to the output space $\mathbf{Y}^s$. Similarly, $H_{\theta,1|L}: \mathbf{X}^{\text{spec}} \to \mathbf{Y}^s$ represents the network H_θ mapping the input $\mathbf{x}^{\text{spec}}$ from the input space $\mathbf{X}^{\text{spec}}$ to the output space $\mathbf{Y}^s$.

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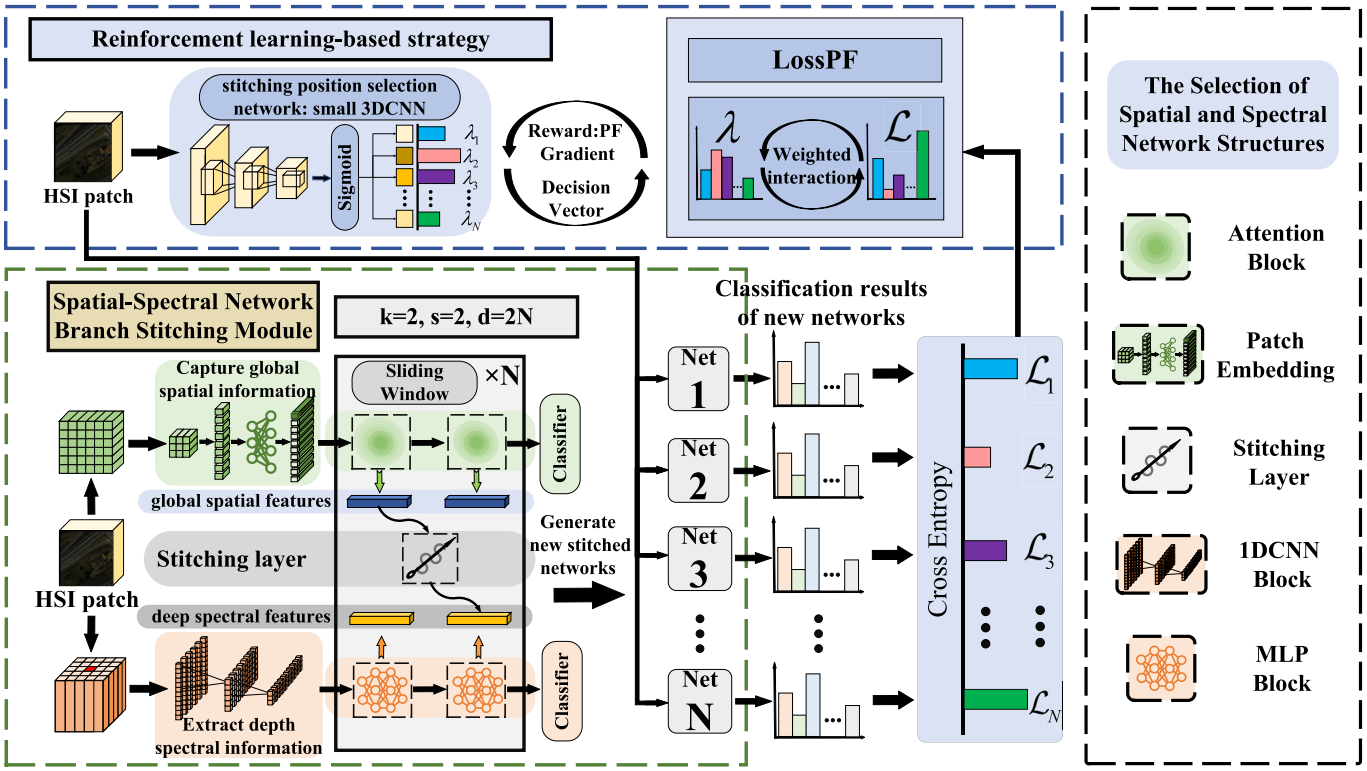


Fig. 2. Overall training pipeline of the proposed DSNet. First, a spatial–spectral network branch stitching module is proposed to insert multiple stitching layers at different positions between the two network branches. Then, an RLBS is designed, and a stitching position selection network is used to assist specific data in choosing the stitching position (subnetwork) to pass through. Finally, considering that the ViT can effectively learn global spatial information, while 1DCNN excels at capturing deep spectral information, these two foundational networks are selected to learn spatial and spectral knowledge, respectively, and are then stitched together.

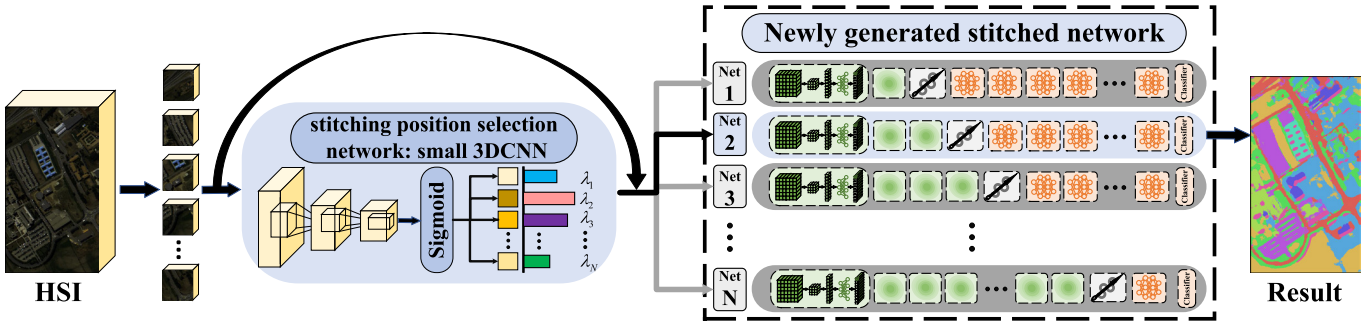


Fig. 3. Overall test workflow of the proposed DSNet. First, the input data is fed into the stitching position selection network to obtain a decision vector. Then, the decision vector is used to determine which subnetwork in the stitched super-network to activate. Finally, the original input data is passed through the selected subnetwork, and the output is used as the final classification result.

To stitch these two networks together, the stitching location needs to be fixed first. Suppose that there are two integer parameters $m, n \in [1, L]$, these two networks are split at these positions, forming four network segments: $T_{\phi,1|m} : \mathbf{X}^{\text{spec}} \to \mathbf{A}_{\phi}^{m}$, $T_{\phi,m+1|L} : \mathbf{A}_{\phi}^{m} \to \mathbf{Y}^{s}$, $H_{\theta,1|n} : \mathbf{X}^{\text{spec}} \to \mathbf{A}_{\theta}^{n}$, and $H_{\theta,n+1|L} : \mathbf{A}_{\theta}^{n} \to \mathbf{Y}^{s}$. $\mathbf{A}_{\phi}^{m}$ and $\mathbf{A}_{\theta}^{n}$ represent the intermediate feature activation spaces of the two networks.

Assume that a stitching layer $S_{m,n}$ can map intermediate features from feature space $\mathbf{A}_{\phi}^{m}$ to feature space $\mathbf{A}_{\theta}^{n}$, it means that the stitching layer can stitch $T_{\phi,1|m}$ and $H_{\theta,n+1|L}$ together to form a new stitched network, which can be mathematically represented as

$$F^{S}\left(\mathbf{X}^{\text{spa}}\right) = H_{\theta,n+1|L} \circ S_{m,n} \circ T_{\phi,1|m}\left(\mathbf{X}^{\text{spa}}\right) = \mathbf{Y}^{s}. \quad (2)$$

By controlling the two parameters m and n according to certain rules, multiple stitching layers can be inserted between the two networks simultaneously, thereby generating a stitched super-network. Assuming that N stitching layers are inserted, the resultant super-network will contain N newly created stitching subnetworks. This process can be mathematically expressed as

$$\mathbf{SupNet}(\mathbf{X}) = \mathbf{Snnet}\langle T, H\rangle(\mathbf{X}) = \left\{F_1^S(\mathbf{X}), \ldots, F_N^S(\mathbf{X})\right\} \quad (3)$$

where $\mathbf{SupNet}(\mathbf{X})$ represents the overall stitched super-network and $F_i^S(\mathbf{X})$ represents the ith subnetwork contained in the super-network.

Specifically, this module utilizes a sliding window approach to regularly insert stitching layers between two networks.

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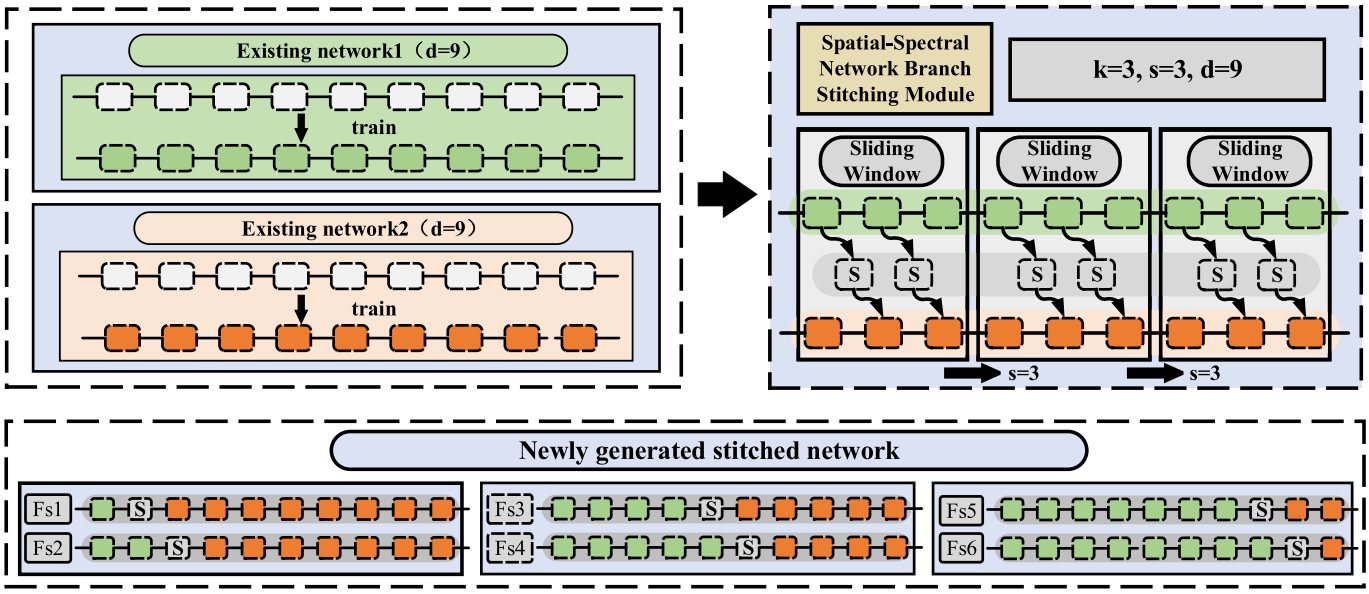


Fig. 4. Given parameter $d = 9$, $k = 3$, and $s = 3$ as an example to show the workflow of the spatial–spectral network branch stitching module. First, two preexisting networks are divided into 9 layers, are trained separately. Then, the sliding window includes $k = 3$ network layers. In the sliding window, the nth layer of one network is stitched to the $n+1$th layer of the other network, inserting $k-1 = 2$ stitching layers. The sliding window then moves $s = 3$ layers toward the end of the network and continues stitching until it has traversed all the network layers. Finally, the two pretrained networks are combined by using the spatial–spectral network branch stitching module to form 6 new networks.

The number of divided network layers can be defined as the network depth d (where $d = L$), and a sliding window is defined for fixing the stitching position. Let k represent the number of network layers contained in the window, which represents the kernel size of the window, and let s represent the number of network layers the window moves at each step, which represents the stride of the window.

During the stitching process, a sliding window is first placed at the starting points of the two networks. In this sliding window, the network layers of T_ϕ and H_θ are stitched together. Specifically, the nth layer of one original network is concatenated to the $(n+1)$th layer of the other original network, which is equivalent to inserting $k-1$ concatenation layers in each sliding window. Once all the network layers in the sliding window have been stitched, the window moves s layers toward the end of the networks, and the stitching process continues for the newly entered network layers. This process is repeated until the sliding window reaches the end of the network, completing the entire stitching process.

To prevent redundant stitching, which wastes computational resources, and to ensure that the sliding window does not exceed the end of the network, it is clear that the parameters d, k, and s must satisfy the following constraints:

$$s \geq k-1, \quad s+k \leq d. \tag{4}$$

To illustrate the overall stitching process more clearly, the parameter set $d = 9$, $k = 3$, and $s = 3$ is taken as an example, using the spatial–spectral network branch stitching module (SSNBS) to stitch the spatial information network T_ϕ and the spectral information network H_θ. The overall stitching process is shown in Fig. 3. Obviously, in this process, stitching layers are inserted, generating 6 new stitched networks. This process can be mathematically represented as

$$\begin{aligned}\mathbf{SupNet}\left(\mathbf{X}^{\text{spa}}\right) &= \mathbf{Snnet}\langle T_{\phi,1|9}, H_{\theta,1|9}\rangle\left(\mathbf{X}^{\text{spa}}\right) \\ &= \left\{F_1^S\left(\mathbf{X}^{\text{spa}}\right)F_2^S\left(\mathbf{X}^{\text{spa}}\right), F_3^S\left(\mathbf{X}^{\text{spa}}\right)\right. \\ &\quad \left. F_4^S\left(\mathbf{X}^{\text{spa}}\right), F_5^S\left(\mathbf{X}^{\text{spa}}\right), F_6^S\left(\mathbf{X}^{\text{spa}}\right)\right\}. \end{aligned} \tag{5}$$

As can be seen from Fig. 4, by controlling the total depth d, window size k, and stride s, the complexity of the super-network can be adjusted. Apparently, when the values of d and k increase and the value of s decreases, the structure of the stitched super-network becomes more complex and contains more newly generated stitched networks.

B. RLBS for Selecting Stitching Positions

To assist in selecting the stitching positions for specific data to pass through, an RLBS is designed. Specifically, a stitching position selection network is introduced to assist the specific data in choosing the newly generated stitching positions in the preceding section to pass through. To enhance the accuracy of the decisions made by the stitching position selection network, the overall method employs reinforcement learning techniques during the training process.

During model inference, the HSI patch is first directly input into the stitching position selection network $\boldsymbol{\psi}_p$, and the final output is passed through a sigmoid function. The resulting vector $\boldsymbol{\lambda}$ is treated as the decision vector. The output dimensions are manually designed to match the number of newly generated stitched networks (i.e., the number of newly stitched networks). Therefore, each element of the decision vector corresponds to one of the newly generated stitched networks. Assuming that the stitched super-network contains N newly generated stitched networks, this process can

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be represented as

$$\boldsymbol{\lambda} = \mathbf{Sigmoid}\left(\boldsymbol{\psi}_p\left(\mathbf{x}^{\mathrm{spa}}\right)\right) = \{\lambda_1, \lambda_2, \ldots, \lambda_N\}\left(\boldsymbol{\lambda} \in \mathbb{R}^N\right). \quad (6)$$

Each element in $\boldsymbol{\lambda}$ has a value between 0 and 1, and these values can be treated as confidence scores for selecting the corresponding network. Since each data batch can only be processed through one network during inference, the network corresponding to the element with the highest value is selected in the decision vector, $\lambda_j = \max\{\lambda_1, \lambda_2, \ldots, \lambda_N\}(j \in [1, N])$, and the jth network $F_j^S(\mathbf{X}^{\mathrm{spa}})$ (the new network with the highest confidence) is activated to perform classification predictions on the hyperspectral data.

To ensure the decisions made by the stitching position selection network are more accurate, the overall method will be trained using reinforcement learning techniques. Specifically, during the training process, the original HSI is simultaneously input into both the stitching position selection network and each of the newly stitched networks. Since each stitched network is equipped with the loss function of the original network H_θ at its end, in addition to obtaining the decision vector $\boldsymbol{\lambda}$, a loss vector $\boldsymbol{L}$ is also obtained, which consists of the loss values output by each super-network. This process can be mathematically expressed as

$$\begin{aligned}\boldsymbol{L} &= \mathbf{SupNet}\left(\mathbf{x}^{\mathrm{spa}}\right) = \mathbf{Snnet}\langle T_\phi, H_\theta\rangle\left(\mathbf{x}^{\mathrm{spa}}\right) \\ &= \left\{F_1^S\left(\mathbf{x}^{\mathrm{spa}}\right), \ldots, \ F_N^S\left(\mathbf{x}^{\mathrm{spa}}\right)\right\} = \{L_1, \ldots, L_N\}\left(\boldsymbol{L} \in \mathbb{R}^N\right).\end{aligned} \quad (7)$$

The smaller the value of the element L_i in the loss vector $\boldsymbol{L}$, the more accurate the prediction made by the corresponding ith network. Thus, L_i can be considered as a measure of the accuracy of the ith network $F_i^S(\mathbf{X}^{\mathrm{spa}})$. To improve the performance of all newly stitched networks while simultaneously increasing the accuracy of decisions of the policy network (stitching position selection network), the total loss of the whole model can be calculated as a weighted sum of the accuracies and confidences of the networks. This can be mathematically expressed as

$$\mathcal{L}_{\mathrm{PF}} = \frac{\boldsymbol{\lambda} \cdot \boldsymbol{L}}{\|\boldsymbol{\lambda}\|_1} = \frac{\sum_{i=1}^N \lambda_i \cdot L_i}{\sum_{i=1}^N \lambda_i}. \quad (8)$$

In this way, during the backpropagation process, the model will tend to reduce the values of all elements in the loss vector $\boldsymbol{L}$, allowing each newly stitched network to be fine-tuned. Simultaneously, the model will also tend to increase the decision vector elements (confidence values) corresponding to networks with smaller loss values (better performance), thereby improving the accuracy of the decisions made by the stitching position selection network.

C. Selection of Spatial and Spectral Network Structures

Considering that the ViT network can better capture global information, while the 1DCNN network focuses more on 1-D detailed information, the ViT network structure is chosen as the network T_ϕ for learning spatial information, and the 1DCNN network structure is chosen as the network H_θ for learning spectral information. After training both networks, the main parts of the networks are divided into $d = 6$ layers and use the SSNBS to stitch the two networks together.

Since the role of the stitching layer in the overall model is solely to transfer features from one activation space to another [27], its structure and initialization method should be as simple as possible. In practice, the simplest linear layer was used as the stitching layer, and it was initialized by using the default Kaiming initialization method [41]. The ViT network consists of a patch embedding and attention block for extracting and recording spatial knowledge, while the 1DCNN network consists of a 1DCNN block and an MLP block for extracting and recording spectral knowledge. The structure of each component in the DSNet is shown in Fig. 5.

When the sliding window parameters are set to $k = 2$ and $s = 1$, the maximum number of newly generated stitched networks is achieved. In this case, 5 stitching layers are inserted throughout the process, generating 5 new networks. This process can be expressed as

$$\begin{aligned}&\mathbf{SupNet}\left(\mathbf{X}^{\mathrm{spa}}\right) \\ &= \mathbf{Snnet}\langle \mathrm{ViT}_{1|6}, \mathrm{1DCNN}_{1|6}\rangle\left(\mathbf{X}^{\mathrm{spa}}\right) \\ &= \left\{F_1^S\left(\mathbf{X}^{\mathrm{spa}}\right), F_2^S\left(\mathbf{X}^{\mathrm{spa}}\right), F_3^S\left(\mathbf{X}^{\mathrm{spa}}\right), F_4^S\left(\mathbf{X}^{\mathrm{spa}}\right), F_5^S\left(\mathbf{X}^{\mathrm{spa}}\right)\right\}.\end{aligned} \quad (9)$$

Since hyperspectral datasets are small and exhibit strong similarity, to enhance the overall generalization ability of the model, two common data augmentation techniques, mixup and cutmix, are applied to preprocess the original HSI data.

Considering that the 3DCNN structure can comprehensively learn spectral–spatial information in HSIs, a small five-output 3DCNN network is used as the stitching position selection network $\boldsymbol{\psi}_p$ to learn the relevant knowledge of the spectral–spatial information and generate the decision vector $\boldsymbol{\lambda}_{vc}$. This process can be expressed as

$$\begin{aligned}\boldsymbol{\lambda}_{vc} &= \mathbf{Sigmoid}\left(\mathrm{3DCNN}\left(\mathbf{x}^{\mathrm{spa}}\right)\right) \\ &= \{\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5\}\left(\boldsymbol{\lambda}_{vc} \in \mathbb{R}^5\right).\end{aligned} \quad (10)$$

Simultaneously, during the training process, the original data is also directly fed into the spectral–spatial stitched super-network to obtain the loss vector $\boldsymbol{L}_{vc}$. This process can be expressed as

$$\begin{aligned}\boldsymbol{L}_{vc} &= \mathbf{SupNet}\left(\mathbf{x}^{\mathrm{spa}}\right) = \mathbf{Snnet}\langle \mathrm{ViT}_{1|6}, \mathrm{1DCNN}_{1|6}\rangle\left(\mathbf{x}^{\mathrm{spa}}\right) \\ &= \left\{F_1^S\left(\mathbf{x}^{\mathrm{spa}}\right), \ldots, \ F_5^S\left(\mathbf{x}^{\mathrm{spa}}\right)\right\} \\ &= \{L_1, L_2, L_3, L_4, L_5\}\left(\boldsymbol{L}_{vc} \in \mathbb{R}^5\right).\end{aligned} \quad (11)$$

Each element in the loss vector is calculated using the cross-entropy loss at the tail of the corresponding stitched network (which is also the tail of the original 1DCNN network). The calculation formula is as follows:

$$\mathcal{L}_i = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic}\log\left(\hat{y}_{ic}\right). \quad (12)$$

To improve the performance of the 3DCNN stitching position selection network while fine-tuning each newly stitched network, the total loss $\mathcal{L}_{\mathrm{PF}}$ can be calculated by weighting the

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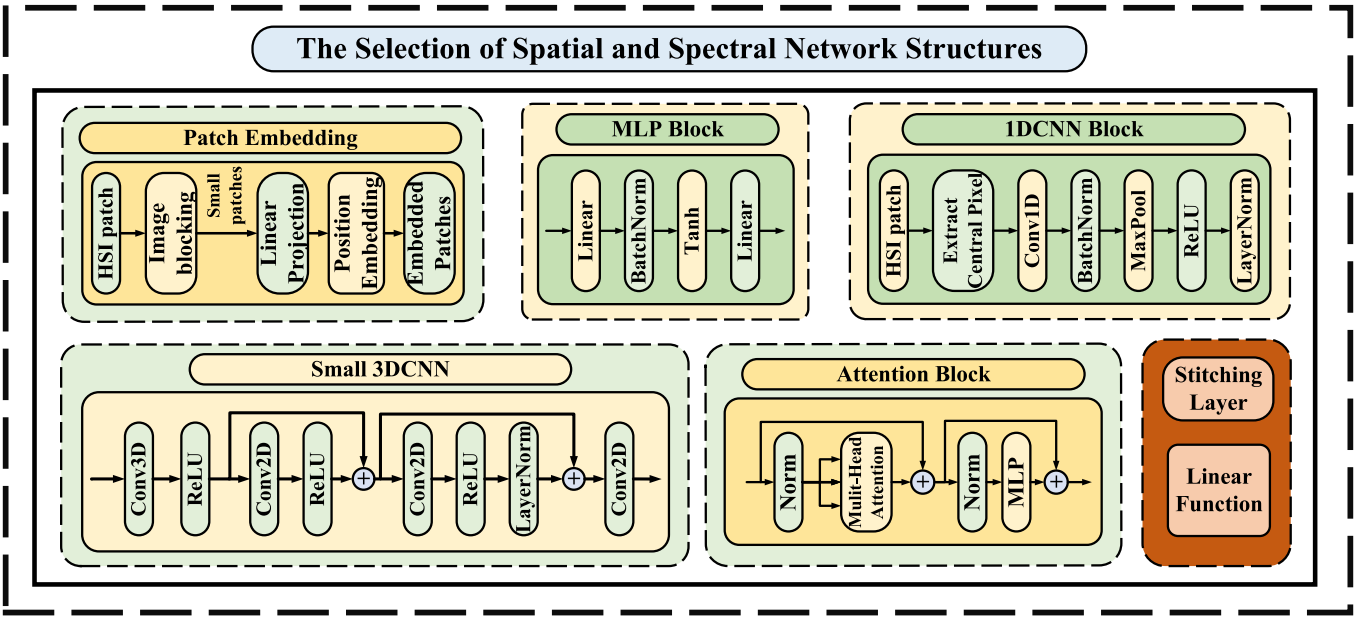


Fig. 5. Structure of each component in the DSNet, including patch embedding, attention block, 1DCNN block, MLP block, small 3DCNN, and stitching layer.

accuracy from the loss vector with the confidence values from the decision vector. The calculation formula is as follows:

$$\mathcal{L}_{\mathrm{PF}} = \frac{\boldsymbol{\lambda}_{vc} \cdot \boldsymbol{L}_{vc}}{\|\boldsymbol{\lambda}_{vc}\|_1} = -\frac{\sum_{i=1}^{5} \lambda_i \sum_N \sum_C y_{ic} \log\left(\hat{y}_{ic}\right)}{N \sum_{i=1}^{5} \lambda_i} \tag{13}$$

where y_{ic} represents the true label and $\hat{y}_{ic}$ represents the predicted label for class c produced by the ith network.

As shown in Fig. 3, the test data is first fed into the stitching position selection network to obtain the decision vector in the test phase. The newly stitched network corresponding to the element with the highest value in the decision vector is activated. The original test data is then input into the activated stitched network, and the resulting output is treated as the final classification result.

IV. Experimental Results and Analysis

In this section, to evaluate the performance of the proposed DSNet, three HSI datasets are used as benchmarks: Houston2013, PaviaU, and YC. Then, the effects of different modules are examined, and the parameters of the DSNet are analyzed. Furthermore, the overall accuracy (OA), average accuracy (AA), and Kappa coefficient are employed as the evaluation metrics for classification performance. Finally, the DSNet is compared against eight HSI classification methods, including SVM [42], 3DCNN [43], 3D–2DCNN feature hierarchy (HybridSN) [44], spectral–spatial residual network (SSRN) [45], double-branch dual-attention mechanism (DABA) network [21], spectral–spatial feature tokenization Transformer (SSFTT) [46], hybrid Transformer (HybridFormer) [47], channel-layer-oriented lightweight spectral–spatial network (CLOLN) [48], lightweight multiscale neural architecture search with spectral–spatial attention (LMSS-NAS) [49], and groupwise separable convolutional ViT (GSC-ViT) [50].

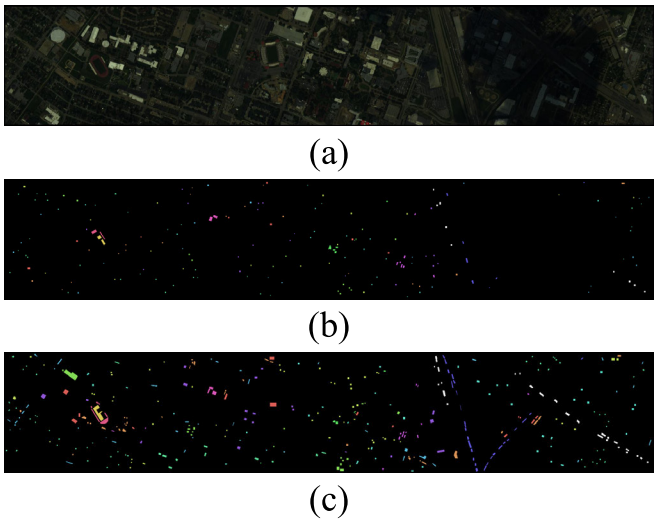


Fig. 6. Houston2013 dataset. (a) False-color image. (b) Training set. (C) Test set.

A. Datasets Description

1) Houston2013 Dataset [51]: The Houston2013 dataset was acquired by the ITRES CASI-1500 sensor over the campus of the University of Houston and adjacent rural areas in Texas, USA.[1] The dataset consists of 349 × 1905 pixels, with 144 spectral bands ranging from 364 to 1046 nm at 10-nm intervals. Table I provides a detailed description of 15 challenging land-cover and land-use classes. The false-color composite and ground-truth maps of the Houston2013 dataset are shown in Fig. 6.

[1]http://www.classic.grss-ieee.org/community/technical-committees/data-fusion/2013-ieee-grss-data-fusion-contest/

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TABLE I
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE HOUSTON2013 DATASET

No.	Color	Land cover type	Training Set	Test Set
1		Healthy Grass	198	1053
2		Stressed Grass	190	11064
3		Synthetic Grass	192	505
4		Tree	188	1056
5		Soil	186	1056
6		Water	182	143
7		Residential	196	1072
8		Commercial	191	1053
9		Road	193	1059
10		Highway	191	1036
11		Railway	181	1054
12		Parking Lot1	192	1041
13		Parking Lot2	184	285
14		Tennis Court	181	247
15		Running Track	187	473
Total			2832	12197

TABLE II
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE PAVIA UNIVERSITY DATASET

No.	Color	Land cover type	Training Set	Test Set
1		Asphalt	548	6304
2		Meadows	540	18146
3		Gravel	392	1815
4		Trees	524	2912
5		Metal Sheets	265	1113
6		Bare Soil	532	4572
7		Bitumen	375	981
8		Bricks	514	3364
9		Shadows	231	795
Total			3921	40002

2) PaviaU Dataset [13]: The Pavia University (PaviaU) dataset[2] was obtained by the reflective optics system imaging spectrometer (ROSIS) sensor over the University of Pavia and its surrounding areas in Italy. The image consists of 610 × 340 pixels and includes 103 spectral bands, with a ground sampling distance (GSD) of 1.3 m. Table II provides a detailed description of the nine land-cover classes included in this dataset. The false-color composite and ground-truth maps of the PaviaU dataset are shown in Fig. 7.

3) YC Dataset [52]: The Yancheng Wetland (YC) dataset was captured on April 4, 2019, by the ZY1-02D AHSI sensor. The spatial dimensions of the YC dataset are 1175 × 585 pixels, with a spatial resolution of 30 m, primarily covering the coastal region of Yancheng, Jiangsu, China. The Yancheng Wetland is rich in resources, featuring vast areas of radiating sandbars, coastal forest farms, and grasslands. After removing the noisy spectral bands, the dataset includes 147 spectral bands in the 0.4–2.5-μm range. Table III provides a detailed description of the 18 land-cover classes included in this dataset. The false-color composite and ground-truth maps of the YC dataset are shown in Fig. 8.

[2]http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

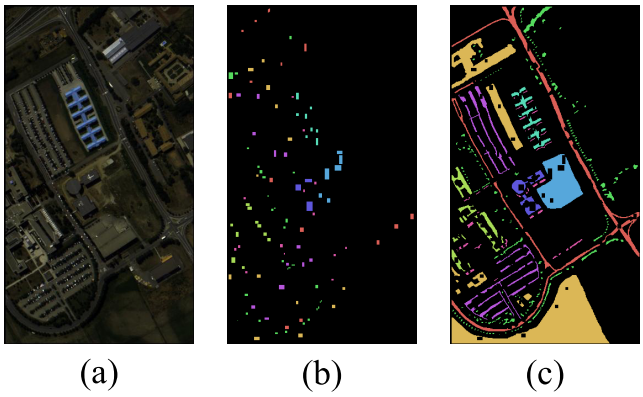


Fig. 7. PaviaU dataset. (a) False-color image. (b) Training set. (C) Test set.

TABLE III
LAND-COVER CLASSES AND THE NUMBER OF SAMPLES FOR THE YANCHENG DATASET

No.	Color	Land cover type	Training Set	Test Set
1		Sea	209	2186
2		Offshore area	140	1448
3		Salt field	6	104
4		Pond	20	173
5		Partina anglica	7	76
6		Mudflats	25	243
7		Aquaculture pond	25	238
8		Paddy filed	87	745
9		Estuarine area	27	248
10		River	27	272
11		Woodland	19	196
12		Barren	25	129
13		Building	37	489
14		Fallow land	26	208
15		Rainfed cropland	28	176
16		Suaeda salsa	14	71
17		Irrigation canal	10	28
18		Phragmites	12	120
Total			744	7150

B. Implementation Details

The proposed DSNet is implemented by using the PyTorch framework. All methods are trained on the same hardware platform, consisting of a GTX 3090 GPU, Intel 4210R CPU, and 90GB of memory. According to the proposed method, two original networks (ViT and 1DCNN) are first trained directly on the dataset. Afterward, the SSNBS is approached to insert some stitching layers between the two domain-trained networks, generating five new networks. Finally, a small 3DCNN

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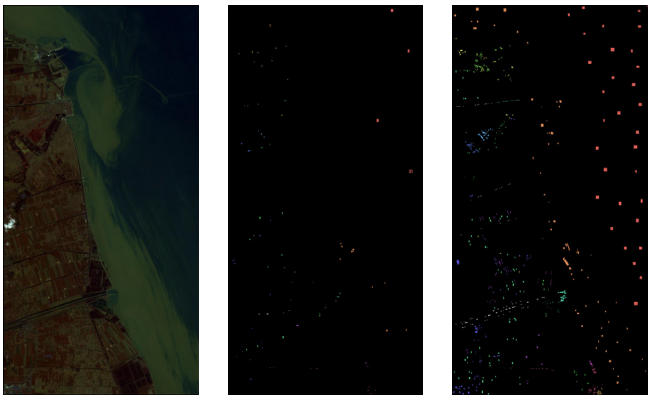


Fig. 8. YC dataset. (a) False-color image. (b) Training set. (C) Test set.

is introduced as the stitching position selection network, and the RLBS is used to train and fine-tune the overall model.

During this process, the inserted stitching layers are implemented as simple linear layers, and both the stitching layers and the stitching position selection network are initialized using random seeds and Kaiming initialization. The networks are divided into six layers in the middle, and the parameters for the sliding window in the stitching process are set to $k = 2$ and $s = 1$ under the initial conditions. The mixup and cutmix data augmentation coefficients are both set to 0.1. The experiments are conducted on three datasets with separate training and test sets [13], and 10% of the training data samples are randomly selected as the validation set for model development. Additionally, the batch size, number of training epochs, learning rate of the Adam optimizer, and weight decay are set to 128, 150, 0.001, and $1e^{-4}$, respectively. The classification performance is compared using three evaluation metrics: OA, AA, and Kappa coefficient. To avoid biased estimates, all experiments with the DSNet are independently tested. In parameter analysis and ablation study, all experimental results are obtained by independently running the experiments five times and then taking the average and variance. Meanwhile, in the comparative experiments, all experimental results are obtained by independently running the experiments ten times and then taking the average and variance.

C. Parameter Analysis

In this section, the impact of three adjustable parameters in the model is analyzed through experimental analysis.

First, the influence of network depth (number of layers) d is discussed on the performance of the DSNet. When the sliding window parameters (kernel size k and stride s) are fixed, changing the network depth d of the two networks affects the number of generated networks, which, in turn, impacts the overall model performance. Let $k = 2$, $s = 1$, and the patch size is set to 8. The experiments are conducted by varying the network depth d (i.e., $\{3, 4, 5, 6, 7\}$), and OA is used as the evaluation metric.

Fig. 9(a) shows the results of the DSNet with different network depths d on the three datasets. As can be clearly seen from Fig. 9(a), the performance of the DSNet initially increases and then decreases as the network depth d increases. The reason for this trend is that increasing d leads to the generation of more new networks, which provides more options for the data, thereby improving performance. However, when d becomes too large, the number of new networks generated becomes excessive, leading to a larger model size and resulting in overfitting, which degrades the performance of the whole model. When the network depth d is set to 6, the DSNet achieves the best performance on all three datasets.

Next, the impact of the sliding window parameters (kernel size k and stride s) is analyzed. During the experiments, it is essential to avoid redundant stitching, which would waste computational resources, and to ensure that the sliding window does not exceed the network layers. Therefore, the relationship between k, s, and d should satisfy the conditions: $s \geq k - 1$, $s + k \leq d$. In the experiments, the network depth d is set to 6, resulting in the generation of five new stitched networks, which are sequentially numbered. Then, let k and s vary and evaluate the performance using OA as the metric. Table IV presents the experimental results across the three datasets. From the table, it is clear that more stitching layers do not necessarily lead to better model performance. When the number of stitching layers is appropriate and the stitched super-network contains several high-performing subnetworks, the overall model performance tends to improve. Notably, when $k = 2$ and $s = 2$, the model achieves the best performance.

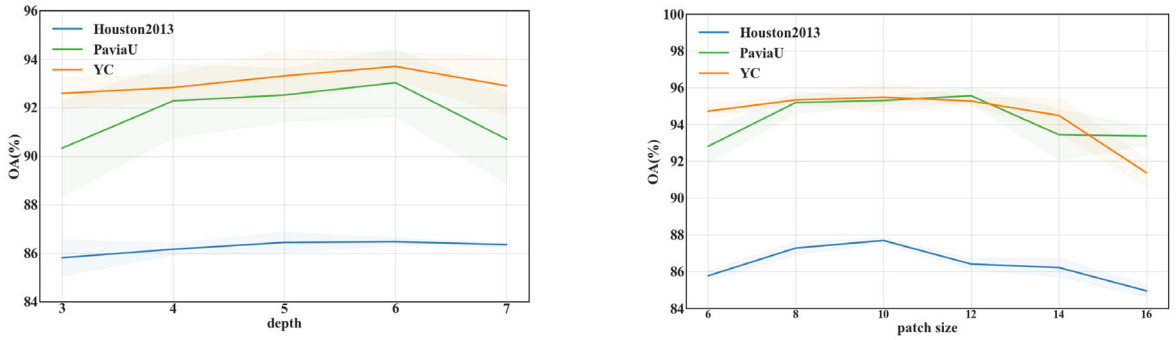
Lastly, the impact of patch size p is analyzed. In HSIC problems, p primarily affects the spatial contour of hyperspectral pixels. Since the stitching position selection network needs to select a subnetwork within the stitched super-network based on the information from the input image patches during training, the size of p directly influences the information learned by the stitching position selection network and the decisions it makes. The candidate range for p is set to {6, 8, 10, 12, 14, 16}, where smaller p values provide fewer but purer spatial features, while larger p values cover a broader receptive field but also introduce more noise. Fig. 9(b) shows the OAs of the DSNet with different patch sizes across the three datasets. As shown clearly in the figure, the OA of the model on all three datasets increases at first and then decreases as the patch size p increases. This is because, within a certain range, a larger patch size provides the model with richer spatial information. However, when the patch size becomes too large, the complexity of land-cover distribution leads to an increased proportion of interfering pixels. Evidently, setting the patch size to ten yields optimal performance for the model on the Houston2013 and PaviaU datasets, while a patch size of 12 achieves the best results on the YC dataset.

D. Ablation Study

In this section, ablation experiments are conducted to validate the contribution of each module in the DSNet to the overall performance. Specifically, Table V presents the performance results of all variations.

For each dataset, the first and second rows of the modules in Table V represent experiments where only the 1DCNN network for spectral information extraction and the ViT network

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(a) (b)

Fig. 9. OA of the DSNet versus different values of d and p. (a) Depth d. (b) Patch size p.

TABLE IV
OA VERSUS THE VALUE OF s AND k ON THREE DATASETS. THE BEST ONE IS SHOWN IN BOLD

$d=6$			OA		
k	s	NewNet	Houston13	PaviaU	YC
2	1	1,2,3,4,5	86.47±0.18	93.03±2.71	92.70±2.21
	2	1,3,5	**87.28±0.46**	**95.19±0.74**	**95.43±0.02**
	3	1,4	86.18±0.60	93.25±1.08	93.61±0.85
	4	1,5	85.85±0.29	92.89±0.74	92.33±0.84
3	2	1,2,3,4	86.77±0.81	92.83±1.89	91.70±1.63
	3	1,2,4,5	85.90±0.46	90.94±2.51	92.01±1.44

TABLE V
ABLATION STUDY ON THREE DATASETS. THE BEST ONE IS SHOWN IN BOLD

Dataset	Modules				Results		
	1DCNN	ViT	SSNBS	RLBS	OA	AA	Kappa
Houston2013	✓	×	×	×	81.42%	84.45%	79.85%
	×	✓	×	×	84.41%	86.85%	83.15%
	✓	✓	✓	×	85.77%	87.59%	84.57%
	✓	✓	✓	✓	**87.21%**	**89.28%**	**86.16%**
PaviaU	✓	×	×	×	80.26%	86.61%	74.79%
	×	✓	×	×	91.89%	90.93%	89.01%
	✓	✓	✓	×	92.10%	92.14%	89.51%
	✓	✓	✓	✓	**95.08%**	**94.41%**	**93.32%**
YC	✓	×	×	×	92.87%	86.40%	90.93%
	×	✓	×	×	93.63%	85.83%	92.46%
	✓	✓	✓	×	93.91%	84.95%	92.77%
	✓	✓	✓	✓	**95.13%**	**86.01%**	**94.21%**

for spatial information extraction were retained, respectively. The third row represents the experiment in which the pre-trained ViT and 1DCNN were stitched using the SSNBS to generate several new networks, and the data was randomly passed through these newly generated networks. The last row represents the experiment where the RLBS was applied, using the stitching position selection network to allocate the newly generated stitched networks for each data sample.

The results clearly indicate that, across all three datasets, the experiments with the stitched super-networks yielded better performance than those with each individual network alone. This phenomenon indicates that the SSNBS effectively stitches and fine-tunes the ViT and 1DCNN networks. The resulting new stitched networks integrate both global spatial knowledge from the ViT and deep spectral knowledge from 1DCNN, leading to enhanced performance.

Furthermore, when the RLBS was introduced, the performance of the model improved across all three datasets. This indicates that the RLBS can effectively allocate the most appropriate model based on the intrinsic information of the data, thereby enhancing the overall performance of the model.

E. Comparison With Advanced HSIC Methods

To evaluate the overall performance of the DSNet, a series of HSIC methods are implemented on the same experimental platform for a fair comparison. These methods include the SVM [42], 3DCNN [43], HybridSN [44], SSRN [45],

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TABLE VI

CLASSIFICATION ACCURACIES OF METHODS ON THE HOUSTON13 DATASET. THE BEST OA AND KAPPA ARE SHOWN IN BOLD

Class	SVM [42]	3DCNN [43]	HybridSN [44]	SSRN [45]	DABA [21]	SSFTT [46]	HybridFormer [47]	CLOLN [48]	LMSS-NAS [49]	GSC-ViT [50]	DSNet
1	**88.82±0.00**	82.18±0.45	82.65±0.27	82.24±0.38	83.69±2.84	77.23±1.88	81.84±0.37	81.92±0.43	80.36±0.33	81.39±0.57	82.65±3.34
2	**90.68±0.00**	90.45±7.04	83.57±0.61	86.50±4.41	85.00±0.08	85.70±5.59	84.64±0.79	87.07±4.93	81.05±1.46	84.68±0.77	84.99±0.27
3	85.45±0.00	92.54±2.92	87.43±4.12	98.16±0.37	98.93±0.41	93.19±0.85	95.09±0.64	98.14±2.42	76.04±3.55	96.20±1.68	**99.41±0.46**
4	94.82±0.00	89.25±1.20	92.02±1.06	95.75±2.86	92.96±0.15	84.66±3.38	92.25±0.55	93.99±2.85	88.75±0.39	90.15±0.70	**97.65±2.94**
5	93.94±0.00	99.19±0.36	99.84±0.17	100.00±0.00	100.00±0.00	100.00±0.00	99.92±0.11	99.98±0.04	99.92±0.11	100.00±0.00	**100.00±0.00**
6	19.95±0.00	86.36±3.51	94.69±1.14	95.66±0.52	96.36±0.81	96.48±2.06	95.39±3.95	96.50±1.47	78.04±1.63	96.08±0.56	**96.57±0.66**
7	72.82±0.00	84.65±2.71	84.47±2.38	87.63±5.09	86.98±3.66	75.15±2.68	81.31 ±1.68	80.49±9.94	74.14±2.21	83.96±1.53	**87.92±1.97**
8	33.41±0.00	44.35±2.52	66.28±2.62	60.67±8.57	55.76±6.82	**86.76±7.51**	71.34±8.86	66.44±8.02	66.27±2.39	68.00±7.08	81.79±5.01
9	69.21±0.00	81.07±1.60	82.78±1.85	83.67±3.00	**89.25±1.15**	86.21±4.87	83.08±1.29	79.51±2.16	77.39±3.42	79.17±1.21	81.18±1.30
10	37.62±0.00	47.88±0.86	46.42±1.67	54.39±6.22	52.30±2.58	58.78±2.08	**59.50±4.87**	58.84±8.22	44.25±2.61	59.21±6.29	56.68±2.69
11	54.64±0.00	66.78±2.59	67.61±0.93	84.49±1.97	**92.47±4.19**	91.27±4.99	84.71±1.01	75.33±2.24	64.33±1.76	84.21±3.97	86.43±0.90
12	34.39±0.00	85.46±2.42	86.55±2.30	94.58±4.54	**98.31±1.14**	98.25±0.69	95.24±2.31	86.65±6.91	84.32±3.86	93.24±2.31	96.67±2.71
13	30.85±0.00	81.12±3.04	88.48±3.24	87.33±1.97	87.37±1.24	83.23±3.04	84.07±1.78	88.12±4.33	85.75±1.67	83.37±3.39	**88.66±2.53**
14	74.12±0.00	91.70±2.79	94.70±4.34	100.00±0.00	100.00±0.00	99.92±0.16	99.84±0.2	97.81±2.57	91.01±3.54	99.76±0.20	**100.00±0.00**
15	99.04±0.00	88.75±5.23	93.45±2.37	99.75±0.31	**99.92±0.17**	97.93±4.04	97.55±2.77	98.35±1.91	98.48±1.49	97.93±2.86	98.87±0.40
OA(%)	67.91±0.00	78.78±0.93	80.89±0.56	84.96±1.25	85.54±0.58	85.74 ±1.01	84.93±0.83	83.15±0.83	77.54±0.55	84.13±0.49	**87.21±0.48**
AA(%)	65.31±0.00	80.78±0.96	83.43±0.72	87.42±0.89	87.95±0.50	87.72±0.81	87.05±0.85	86.01±0.86	79.34±0.70	86.49±0.43	**89.28±0.48**
Kappa	65.50±0.00	77.31±1.06	79.69±1.21	84.58±1.86	84.34±0.63	84.52±1.10	83.72±0.89	81.81±1.21	75.76±0.60	82.86±0.52	**86.16±0.51**

TABLE VII

CLASSIFICATION ACCURACIES OF METHODS ON THE PAVIAU DATASET. THE BEST OA AND KAPPA ARE SHOWN IN BOLD

Class	SVM [42]	3DCNN [43]	HybridSN [44]	SSRN [45]	DABA [21]	SSFTT [46]	HybridFormer [47]	CLOLN [48]	LMSS-NAS [49]	GSC-ViT [50]	DSNet
1	80.44±0.00	85.86±2.33	83.69±1.78	95.90±0.77	94.98±0.51	84.92±1.93	92.85±1.41	90.12±1.66	91.27±0.99	91.92±2.57	**96.95±0.64**
2	77.77±0.00	89.94±3.13	90.68±1.56	80.53±5.81	76.80±2.04	72.88±1.21	90.40±6.09	90.80±3.36	88.53±3.49	94.64±1.54	**97.34±0.50**
3	39.48±0.00	58.85±4.64	63.50±2.44	**87.63±2.71**	83.97±3.90	83.59±3.22	80.49±5.26	82.07±2.90	74.16±3.37	85.48±2.68	82.41±1.84
4	66.84±0.00	97.13±1.02	**98.15±0.29**	97.13±2.33	94.79±1.43	84.36±4.57	95.40±0.58	98.10±0.61	87.12±1.55	94.90±0.58	92.47±1.35
5	96.22±0.00	98.97±0.53	99.17±0.35	**99.77±0.09**	99.48±0.24	99.78±0.19	99.14±0.07	99.50±0.40	99.55±0.39	99.23±0.07	98.47±0.28
6	39.29±0.00	92.17±2.55	75.25±8.56	98.51±0.40	**99.51±0.37**	96.19±2.99	98.36±2.52	96.50±4.42	94.09±2.19	90.99±9.52	84.63±6.60
7	58.73±0.00	87.78±1.86	83.10±4.74	96.37±1.13	98.39±0.25	97.98±1.88	95.05±3.59	97.74±1.61	96.41±2.85	98.45±0.76	**98.93±0.61**
8	77.97±0.00	97.57±0.57	96.80±0.73	97.46±0.48	98.85±0.26	91.13±2.49	98.78±0.21	98.86±0.38	98.67±0.53	98.18±0.71	**99.44±0.16**
9	**99.94±0.00**	94.43±1.04	95.23±0.97	98.51±0.23	95.37±4.17	97.41±1.14	97.51±0.59	97.91±0.95	93.84±0.33	96.76±0.61	99.09±0.21
OA(%)	71.97±0.00	89.59±1.88	87.78±0.80	89.51±2.73	87.30±0.98	82.15±1.42	92.81±2.76	92.71±2.05	90.30±1.41	93.98±1.38	**95.08±0.80**
AA(%)	70.74±0.00	89.18±0.95	87.29±1.20	94.43±0.93	93.68±0.41	89.81±1.38	94.22±1.14	94.62±0.98	91.52±0.17	94.62±1.04	**94.71±1.03**
Kappa	63.20±0.00	86.48±2.65	84.15±1.37	86.71±3.53	83.64±1.19	77.17±1.79	90.53±3.49	90.35±2.66	87.17±1.76	91.93±1.88	**93.32±1.11**

TABLE VIII

CLASSIFICATION ACCURACIES OF METHODS ON THE YC DATASET. THE BEST OA AND KAPPA ARE SHOWN IN BOLD

Class	SVM [42]	3DCNN [43]	HybridSN [44]	SSRN [45]	DABA [21]	SSFTT [46]	HybridFormer [47]	CLOLN [48]	LMSS-NAS [49]	GSC-ViT [50]	DSNet
1	98.41±0.00	87.69±7.64	77.16±3.90	88.72±6.14	89.54±3.20	87.83±3.40	99.20±1.41	91.80±8.35	78.78±7.97	95.30±8.12	**99.30±0.46**
2	100.00±0.00	100.00±0.00	100.00±0.00	99.69±0.44	100.00±0.00	98.85±1.24	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
3	5.17±0.00	17.89±5.44	56.64±13.68	67.31±6.92	**75.19±2.75**	24.62±14.82	46.54±15.19	51.35±21.66	63.27±18.28	46.54±20.39	37.89±29.19
4	0.00±0.00	72.71±2.24	81.04±3.59	89.13±5.78	86.01±4.41	78.38±2.27	80.69±1.30	81.50±3.99	28.32±6.48	**89.48±2.91**	81.97±5.51
5	0.00±0.00	38.95±6.12	72.76±15.36	97.11±5.82	**99.21±1.58**	84.47±21.90	43.68±11.69	84.74±17.82	82.63±13.86	72.37±24.00	92.90±8.79
6	78.95±0.00	83.87±5.83	87.28±3.55	92.88±3.11	93.09±0.80	95.23±1.12	92.18±2.69	92.43±2.12	87.16±4.89	**96.79±1.87**	94.65±4.38
7	60.10±0.00	87.14±6.69	91.77±1.52	89.96±8.98	93.36±2.12	92.89±1.21	**99.66±0.41**	92.86±4.00	52.61±3.23	96.39±1.65	96.47±2.00
8	92.49±0.00	98.19±0.49	97.11±0.62	94.67±4.03	99.25±0.37	88.97±3.95	99.14±0.28	99.22±0.92	97.93±0.84	99.20±0.62	**99.60±4.38**
9	88.07±0.00	93.11±2.50	98.99±1.04	99.15±0.77	**99.92±0.16**	94.44±3.02	99.19±1.42	99.52±0.47	99.44±0.60	95.73±2.05	96.53±2.45
10	38.73±0.00	96.51±1.92	**98.36±0.46**	90.29±6.40	97.06±2.21	54.34±10.74	91.10±0.85	95.00±4.25	86.77±7.85	95.52±4.16	83.75±11.03
11	100.00±0.00	94.34±4.96	100.00±0.00	100.00±0.00	99.18±1.15	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	100.00±0.00	**100.00±0.00**
12	60.42±0.00	57.36±8.02	96.90±2.48	**99.54±1.16**	98.30±0.90	94.88±1.60	95.35±2.14	97.36±0.62	99.49±0.38	97.83±1.58	96.28±1.86
13	65.37±0.00	78.59±4.93	84.23±5.93	94.44±5.04	92.84±6.71	96.03±3.39	92.23±6.39	82.33±9.29	85.32±5.42	95.71±3.49	**98.14±2.27**
14	83.30±0.00	97.40±2.93	**98.46±1.00**	82.98±5.32	80.67±4.46	85.10±6.87	91.25±2.32	87.59±2.05	92.02±3.26	90.48±3.43	78.85±6.33
15	66.12±0.00	**96.31±1.87**	90.80±4.57	80.51±6.82	75.91±5.42	64.66±11.72	75.23±3.78	88.64±5.12	74.21±3.88	79.21±5.23	89.55±4.04
16	31.82±0.00	73.10±5.13	73.24±5.45	78.31±4.76	78.59±5.80	**87.32±6.84**	65.07±1.64	80.85±9.20	79.16±7.31	69.86±5.39	79.44±7.05
17	12.50±0.00	80.36±14.13	**98.21±1.79**	70.71±14.27	90.00±6.92	30.00±17.29	82.14±5.98	96.43±7.14	91.43±10.74	82.14±13.92	67.86±6.39
18	0.00±0.00	55.25±2.77	73.08±9.09	67.08±12.41	48.00±3.32	53.17±5.90	57.50±1.90	73.33±12.42	**77.67±8.70**	60.17±8.60	55.17±3.18
OA(%)	84.73±0.00	88.54±2.51	88.40±1.14	91.94±2.27	92.98±0.47	87.93±1.11	94.57±0.72	92.84±2.46	85.82±2.24	94.44±2.59	**95.13±1.00**
AA(%)	54.52±0.00	78.26±1.00	87.55±1.04	87.92±1.90	**89.86±1.03**	78.39±1.97	83.90±0.92	88.59±1.76	82.02±1.66	86.82±1.38	86.01±2.90
Kappa	81.84±0.00	87.24±3.06	87.19±1.00	91.64±3.32	91.74±0.54	85.79±1.31	93.55±0.84	91.58±2.83	83.38±2.51	93.45±2.98	**94.21±1.18**

DABA [21], SSFTT [46], HybridFormer [47], CLOLN [48], LMSS-NAS [49], and GSC-ViT [50]. All methods are implemented on the same hardware platform, with all corresponding hyperparameters consistent with the optimal configurations in the original articles. The results of all the comparative algorithms on the Houston2013, PaviaU, and YC datasets are presented in Tables VI–VIII. Additionally, Figs. 10–12 show the ground-truth maps and the corresponding classification maps for three datasets. It is worth noting that the individual dataset used for evaluation presents greater challenges compared to random sampling. This is because the training and test data are disjoint and do not overlap. Evidently, compared to all other methods, the DSNet shows the best performance across all three datasets. Among all other methods, CLOLN and LMSS-NAS are, respectively, based on dual-branch and NAS methods. It is worth noting that the DSNet outperforms both CLOLN and LMSS-NAS on all three datasets, indicating the superiority of the DSNet over traditional dual-branch and NAS methods.

It is well known that the Houston dataset poses challenges to HSIC methods, primarily due to the significant differences between the training and test samples [13]. However, the DSNet still ranks first in seven categories of the Houston dataset, and it achieves the highest OA, AA, and Kappa. From the classification maps, the DSNet produces clearer and smoother results compared to other methods. These observations suggest that our approach exhibits a certain degree of robustness to data. It can be inferred that even if the

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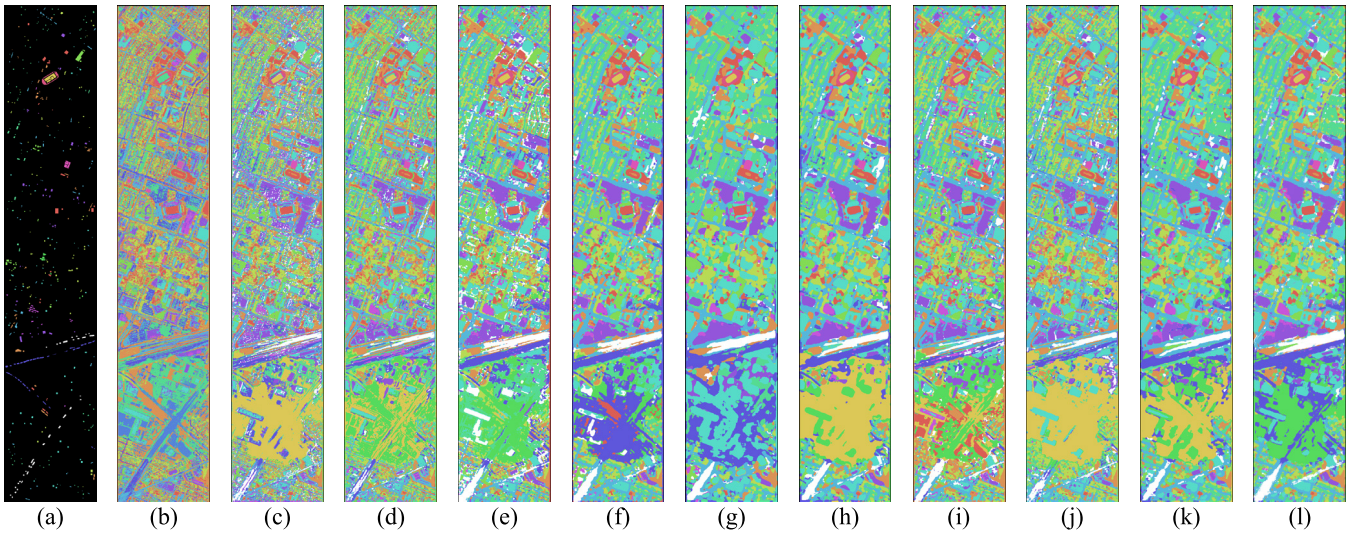


Fig. 10. Classification maps of different methods on the Houston13 dataset. (a) Ground truth. (b) SVM. (c) 3DCNN. (d) HybridSN. (e) SSRN. (f) DABA. (g) SSFTT. (h) HybridFormer. (i) CLOLN. (j) LMSS-NAS. (k) GSC-ViT. (l) DSNet.

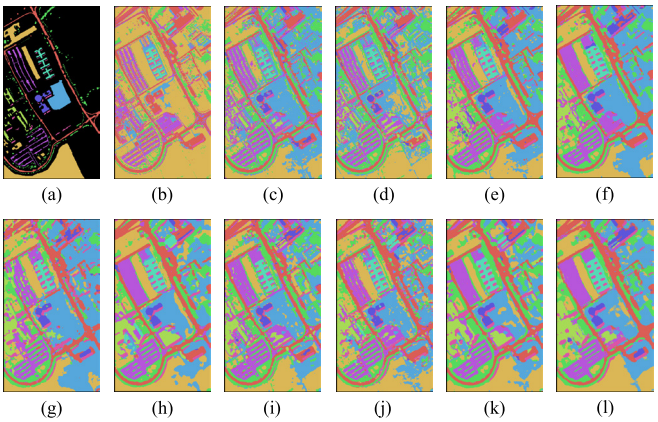


Fig. 11. Classification maps of different methods on the PaviaU dataset. (a) Ground truth. (b) SVM. (c) 3DCNN. (d) HybridSN. (e) SSRN. (f) DABA. (g) SSFTT. (h) HybridFormer. (i) CLOLN. (j) LMSS-NAS. (k) GSC-ViT. (l) DSNet.

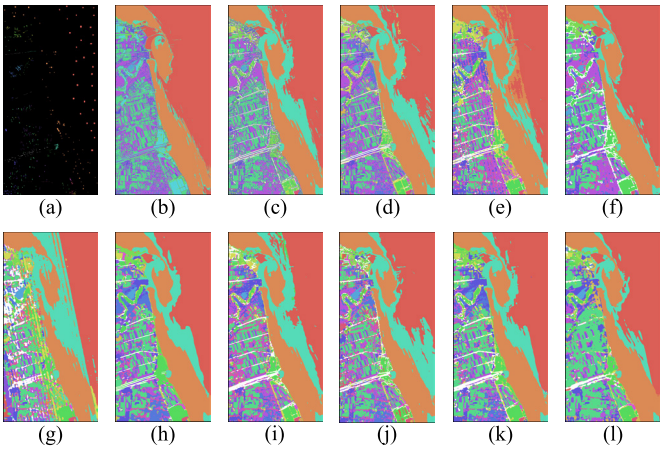


Fig. 12. Classification maps of different methods on the YC dataset. (a) Ground truth. (b) SVM. (c) 3DCNN. (d) HybridSN. (e) SSRN. (f) DABA. (g) SSFTT. (h) HybridFormer. (i) CLOLN. (j) LMSS-NAS. (k) GSC-ViT. (l) DSNet.

sample distribution is complex and has significant differences, the DSNet is still able to effectively learn the information embedded in the dataset.

For the PaviaU dataset, there is a significant imbalance in the distribution of different categories. Nevertheless, the DSNet still achieves the best results in classifying most categories. As shown in the table, the DSNet not only shows strong performance in four categories but also ranks first in OA, AA, and the Kappa across all methods. Unsurprisingly, the DSNet produces clearer and more accurate results for most categories in the classification map. This indicates that even when the class distribution in the dataset is imbalanced, our method can still effectively learn the characteristics of the dataset and achieve excellent classification results.

The YC dataset, with the largest number of categories, poses significant classification challenges for the model. As can be observed from the table, among all the methods, the DSNet does not achieve the best AA on the YC dataset. This is primarily due to its suboptimal performance on the salt field category, which significantly drags down the overall AA. The proposed DSNet features a flexible network architecture and strong generalization capabilities, which require a sufficiently large amount of training data to achieve good classification performance. Consequently, on the YC dataset, which contains a large number of categories and has limited and unevenly distributed data for the salt field category, the DSNet tends to misclassify samples from this category into other classes. However, the DSNet still ranks first in five categories and achieves the best OA and Kappa across all methods, which indicates that the DSNet still performs well on the YC dataset. From the classification map, despite the complexity of land-cover distribution in the coastal areas and the large number of categories in the YC dataset, the DSNet still produces high-quality classification maps, and it achieved the best results across all three datasets.

Moreover, to comprehensively evaluate the effectiveness and limitations of our proposed method, a comparative analysis of the computational complexity is conducted across different

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TABLE IX
PARAMETERS, FLOPS, AND COMPUTATIONAL TIME (IN SECONDS) ON EACH DATASET WITH DIFFERENT METHODS

Dataset	Methods	3DCNN [43]	HybridSN [44]	SSRN [45]	DABA [21]	SSFTT [46]	HybridFormer [47]	CLOLN [48]	LMSS-NAS [49]	GSC-ViT [50]	DSNet
Houston2013	#params	0.07M	0.61M	0.10M	0.11M	0.08M	1.33M	**0.01M**	0.53M	**0**.06M	5.24M
	FLOPs	3.10G	2.77G	2.18G	7.20G	0.63G	2.77G	**0.32G**	3.99G	1.09G	7.95G
	Training time	0.63s	0.64s	1.22s	1.27s	0.97s	2.96s	**0.33s**	0.52s	1.62s	3.23s
	Test time	**12.96s**	15.75s	26.81s	30.02s	24.85s	88.87s	18.86s	35.01s	49.73s	96.92s
PaviaU	#params	0.07M	0.61M	0.10M	0.11M	0.08M	1.33M	**0.01M**	0.50M	0.06M	5.24M
	FLOPs	3.10G	2.77G	2.18G	7.20G	0.63G	2.77G	**0.32G**	3.69G	1.09G	6.64G
	Training time	**0.83s**	0.86s	1.73s	1.54s	1.35s	3.93s	4.12s	3.98s	2.32s	4.39s
	Test time	**4.38s**	4.41s	7.89s	9.84s	7.76s	29.89s	4.66s	10.19s	15.62s	29.71s
YC	#params	0.07M	0.61M	0.10M	0.11M	0.08M	1.33M	**0.01M**	0.61M	0.06M	5.24M
	FLOPs	3.10G	2.77G	2.18G	7.20G	0.63G	2.77G	**0.32G**	4.52G	1.09G	5.35G
	Training time	**0.17s**	0.18s	0.35s	0.40s	0.26s	0.81s	0.84s	0.95s	0.47s	0.98s
	Test time	**13.62s**	14.24s	25.86s	33.02s	25.11s	95.01s	29.59s	58.92s	51.61s	78.32s

approaches. Table IX summarizes the number of parameters, floating-point operations (FLOPs), per-epoch training time, and test time of different methods across three datasets. From the table, it can be observed that the DSNet has the largest number of parameters among all methods. This is primarily due to its architecture, which consists of a stitching position selection network and multiple stitched networks, making the overall network structure more complex compared to other methods. Additionally, the DSNet exhibits the longest training time across all three datasets. The reason for this phenomenon is that during training, the data flow must pass through all components of the DSNet simultaneously to ensure that both the stitched position selection network and all stitched networks are optimized. However, the DSNet does not consistently achieve the highest FLOPs or test time among all methods across three datasets. This is attributed to the inference mechanism: during inference, the data flow is directed to the most suitable stitching network after passing through the stitching position selection network, rather than traversing all stitched networks. As a result, many parameters in the DSNet do not participate in the computation during inference, leading to relatively shorter test times. In summary, it can be seen that all indicators of the DSNet are relatively high. This indicates that although our model has high generalization and effectiveness, it comes at the cost of relatively high computational complexity.

V. CONCLUSION

In this article, a novel DSNet is proposed for HSIC. The DSNet adopts the dual-branch structure in the HSIC field and directly stitches the spatial and spectral network branches together by using stitching network methods. Specifically, a spatial–spectral network branch stitching module is proposed to insert multiple stitching layers at different positions between the two network branches, thereby generating a stitched super-network. Each newly generated stitched subnetwork in the stitched super-network incorporates the advantageous parts of both original network branches, leading to excellent performance. Then, to help specific input select the most suitable stitching position in the stitched super-network, an RLBS is designed, and a stitching position selection network is used to assist specific data in choosing the stitching position (subnetwork) to pass through. Finally, considering that the initial part of ViT can effectively learn global spatial information from HSIs, while the terminal part of 1DCNN excels at capturing deep spectral information, the DSNet directly stitches these two networks together. This approach fully leverages the strengths of foundational networks and achieves impressive performance while also saving researchers time and effort in developing new networks. Extensive experiments and analysis on three benchmark HSI datasets show the effectiveness of the DSNet in HSIC tasks.

In the future, we will attempt to use more innovative neural network stitching methods to stitch more existing and advanced feature extractors to make more attempts on different datasets.

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